Cost functions: Point Contact and IK-rigid

1 IK-rigid

$$w_1 \sum_{i=0}^{n} ||({}^{w}_{T}G_i - {}^{w}_{T}F_i)||_2^2 + w_2 \sum_{t=0}^{T-1} \sum_{i=1}^{n} ||({}^{w}_{t}T_0 \cdot {}^{0}_{t}T_X \cdot {}^{X}_{t}H_i - {}^{w}_{t}F_i)||_2^2$$

2 Point contact

$$w_1 \sum_{i=0}^{n} ||({}^{w}_{T}G_i - {}^{w}_{T}F_i).\alpha||_2^2 + w_2 \sum_{t=0}^{T-1} \sum_{i=0}^{n} ||({}^{w}_{t}G_i - {}^{w}_{t}F_i) \cdot \alpha||_2^2$$

where way-points ${}^{w}_{t}G_X$ are generated by linear interpolation of desired pose. The weights are chosen as $w_1 = 1000, w_2 = 10$ and $\alpha = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$. Since point contact with friction assumes that the orientation changes at the contact points do not affect the object pose, the cost function is only on finger-tip positions. We assume that the friction between the foam tips of the Allegro hand and the objects to be sufficiently large.